Q1. Since is irreducible and recurrent

$\exists$ infinite n, $P_{ii}^n = 1$. $\exists m$, $P_{ij}^m > 0$

In the long term, every t times, Marcov chain reenter i, there is a positive probability to enter j.

Assume after k times of visting i, still haven't visit j

$P_{(k)} = (1 - P_{ij}^m)^k$ $\lim_{k\to\infty} (1 - P_{ij}^m)^k = 0$ since $P_{ij}^m > 0$

Thus, there must be at least once. enter j. $f_{ij} = 1$

Q2.

a) $S = \{0, 1, \dots n\}$

$P_{i,i+1} = \lambda(1-\mu)$ $P_{i,i-1} = \mu(1-\lambda)$ [$i > 0$]

$P_{ii} = (1-\lambda)(1-\mu) + \lambda\mu$

$\forall i, j$. P_{ij}^n is combination of steps $+1, -1, 0$.
and $P_{i,i+1}, P_{i,i-1}, P_{ii} > 0$
$\{X_n\}$ is irreducible DTMC

$$(P_{ij}) = \begin{cases} \lambda(1-\mu) & j = i+1 \\ \mu(1-\lambda) & j = i-1 \ \& \ i > 0 \\ (1-\lambda)(1-\mu) + \lambda\mu & j = i \neq 0 \\ 1 - \lambda + \lambda\mu & j = i = 0 \\ 0 & \text{otherwise.} \end{cases}$$

$$\begin{bmatrix} 1-\lambda+\lambda\mu & \lambda(1-\mu) & 0 & \cdots & 0 \\ \mu(1-\lambda) & (1-\lambda)(1-\mu)+\lambda\mu & \lambda(1-\mu) & \cdots & 0 \\ 0 & \mu(1-\lambda) & (1-\lambda)(1-\mu)+\lambda\mu & \cdots & \vdots \\ \vdots & & & & \vdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}$$

b) $S = \{0, 1, \dots m\}$

Probability of change of inventory level happens

$\lambda + \mu \cdot k \qquad k = \min\{1, i\}$

$P(i, i+1) = \frac{\lambda}{\lambda + \mu k} \qquad P(i, i-1) = \frac{\mu k}{\lambda + \mu k}$

$\forall i, j, \; P_{ij}^{n}$ is combination of steps $+1, -1$

$P(i,i+1) \cdot P(i,i-1)^{n} > 0$, $\forall n$ is irreducible DTMC

$$(P_{ij}) = \begin{cases} \frac{\lambda}{\lambda + \mu k} & \text{for } j = i+1 \\ \frac{\mu k}{\lambda + \mu k} & \text{for } j = i-1 \\ 0 & \text{otherwise} \end{cases} \qquad k = \min\{1, i\}$$

$$P = \begin{pmatrix} 0 & 1 & 0 & \cdots & \cdots \\ \frac{\mu}{\lambda+\mu} & 0 & \frac{\lambda}{\lambda+\mu} & \cdots & \\ 0 & \frac{\mu}{\lambda+\mu} & 0 & \frac{\lambda}{\lambda+\mu} & \cdots \\ \vdots & & & & \vdots \end{pmatrix}$$

c) for state 0: $P_{01} = 1$

for $i > 0$ p of increasing $P = \frac{\lambda}{\lambda+\mu}$.

P of decreasing $q = \frac{\mu}{\lambda+\mu}$

① $p = q$. $\lambda = \mu$. , is symmetric ramdom walk.

chain is null recurrent with infinite expected return time

② $p > q$, $\lambda > \mu$ upward drift, transient

③ $p < q$, $\lambda < \mu$. negative drift with bound 0, positive recurrent

d) $p = \frac{\lambda(1-\mu)}{\lambda(1-\mu)+\mu(1-\lambda)}$ $q = \frac{\mu(1-\lambda)}{\lambda(1-\mu)+\mu(1-\lambda)}$

① $p = q$, $\lambda = \mu$, null recurrent

② $p > q$, $\lambda > \mu$ transient

③ $p < q$, $\lambda < \mu$ positive recurrent

e)

Since $\{X_n\}$ is positive recurrent, it is time reversible

$$\pi_i \lambda(1-\mu) = \pi_{i+1} \mu(1-\lambda) \qquad \frac{\pi_{i+1}}{\pi_i} = \frac{\lambda(1-\mu)}{\mu(1-\lambda)} = q$$

$$\pi_1 = \pi_0 q \quad \cdots \quad \pi_n = \pi_0 q^n$$

Since $\pi_0 + \pi_1 + \dots + \pi_n = 1$ once $n \to \infty$, $\pi_0 \frac{1}{1-q} = 1$, $\pi_0 = 1-q$

$$\pi_i = (1-q)q^i$$

$$= \left(1 - \frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right)\left(\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right)^i$$

T3. a) for $j \neq 0, 1$, $q_j(1-p) + q_j p + q_j = 1$,

$$\begin{cases} P_{00} = P_{NN} = 1 \\ P_{i,i+1} = q_i p \\ P_{i,i-1} = q_i(1-p) \\ P_{ii} = 1 - q_i \end{cases}$$

let P_i = P(start from i will reach N before return to 0)

$$P_i = q_i p P_{i+1} + q_i(1-p) P_{i-1} + (1-q_i) P_i$$

$$(q_i p + q_i(1-p)) P_i = q_i p P_{i+1} + q_i(1-p) P_{i-1}$$

$$p P_i + (1-p) P_i = p P_{i+1} + (1-p) P_{i-1}$$

$$P_i = p P_{i+1} + (1-p) P_{i-1}$$

is the form of gambling ruin model. $P_N = 1$

$$P_i = \begin{cases} \dfrac{1-(\frac{1}{p}-1)^i}{1-(\frac{1}{p}-1)^N} & p \neq \frac{1}{2} \\ i/N & p = \frac{1}{2} \end{cases}$$

When first step is counter clockwise, we can treat it as with $q_i(1-p)$ moving forward and $q_i p$ moving backwards,

also, absorbing state change to $P_{00} = 1$, $P_{n+1-i,\,n+1-i} = 1$

$$P'_i = \begin{cases} \dfrac{1-(\frac{p}{1-p})^i}{1-(\frac{p}{1-p})^{n+1-N}} & \text{if } p \neq \frac{1}{2} \\ i/n+1-N & \text{if } p = \frac{1}{2} \end{cases}$$

Thus, $P = q_0 p P_1 + q_0(1-p) P_1'$

$$= \begin{cases} q_0 p \dfrac{1-(\frac{1}{p}-1)}{1-(\frac{1}{p}-1)^i} + q_0(1-p)\dfrac{1-(\frac{p}{1-p})}{1-(\frac{p}{1-p})^{n+1-i}} & \text{if } p \neq \frac{1}{2} \\ q_0 p \frac{1}{i} + q_0(1-p)\frac{1}{n+1-i} & \text{if } p = \frac{1}{2} \end{cases}$$

b) inspects all stations:

$$P = \begin{cases} q_0 p \dfrac{1-(\frac{1}{p}-1)}{1-(\frac{1}{p}-1)^n} + q_0(1-p)\dfrac{1-(\frac{p}{1-p})}{1-(\frac{p}{1-p})^n} & \text{if } p \neq \frac{1}{2} \\ q_0 \frac{1}{n} & \text{if } p = \frac{1}{2} \end{cases}$$

c) A =: inspects all stations before 0
B =: $T_1 < T_2 \dots < T_n$

m_{ij} = P(start from i, reach j before return to 0 or visit any $k > i$ 比起j大)

$$m_{i,i+1} = q_i p + (1-q_i) m_{i,i+1} + q_i(1-p) m_{i-1,i+1}$$

$$m_{i,i+1} = p + (1-p) m_{i-1,i+1}$$

Since can't back to 0, start from i-1, must visit i before i+1

$$m_{i-1,i+1} = 0$$

$$m_{i,i+1} = p$$

$$P(A \cap B) = m_{12} m_{23} \cdots m_{n-1,n} = (p)^{n-1}$$

$$P(B|A) = \begin{cases} p^{n-1} \dfrac{n}{q_0} & \text{if } p = \frac{1}{2} \\ p^{n-1}\left[q_0 p \dfrac{1-(\frac{1}{p}-1)}{1-(\frac{1}{p}-1)^n} + q_0(1-p)\dfrac{1-(\frac{p}{1-p})}{1-(\frac{p}{1-p})^n} \right]^{-1} & \text{if } p \neq \frac{1}{2} \end{cases}$$

T4.

#4 **(Chapter 4, Exercise 64)** Consider a branching process having $\mu < 1$. If $X_0 = n$, calculate the expected number of individuals that ever exist in this population.

$$E\left[\sum_{k=0}^{\infty} X_k \mid X_0 = n\right] = \sum_{k=0}^{\infty} E\left[X_k \mid X_0 = n\right]$$

$$= \sum_{k=0}^{\infty} n\mu^k = \frac{n}{1-\mu}$$

c) $\sum_{n=1}^{\infty} p_{ii}^{n}$:

let $p = \frac{\lambda}{\lambda+\mu}$, $q = \frac{\mu}{\lambda+\mu}$

$$p_{ii}^{n} \ (i \neq 0) = \begin{cases} 0 & \text{odd } n \\ \frac{(2k)!}{k!k!} p^k (1-p)^k & n = 2k \end{cases}$$

$$\sum_{n=1}^{\infty} p_{ii}^{n} = \sum_{k=1}^{\infty} \frac{(2k)!}{k!\,k!} p^k (1-p)^k = \sum_{k=1}^{\infty} \frac{1}{\sqrt{\pi k}} 2^{2k} p^k (1-p)^k$$

$$= \sum_{k=1}^{\infty} \frac{1}{\sqrt{\pi k}} (4p(1-p))^k$$

If $\lambda = \mu$, $p = q = \frac{1}{2}$ $\sum_{n=1}^{\infty} p_{ii}^{n} = \sum_{n=1}^{\infty} \frac{1}{\pi k} = \infty$ recurrence

$\lambda \neq \mu$, $p \neq \frac{1}{2}$ $\sum_{n=1}^{\infty} p_{ii}^{n} < \infty$, transicent

for $i = 0$: $p_{00}^{n} = p_{11} p_{10} = (1-p) p_{11}$ is same as p_{11}

$p_{i,i+1} = \lambda(1-\mu) = pa$ $\quad p_{i,i-1} = (1-\lambda)\mu = pb$

d)

$p_{ii} = (1-\lambda)(1-\mu) + \lambda\mu = pc$

$\sum_{n=1}^{\infty} p_{ii}^{n}$ $\quad p_{ii}^{n} = \sum_{k=0}^{n/2} \frac{n!}{k!\,k!\,(n-2k)!} \cdot pa^k\, pb^k\, pc^{n-2k}$

$$\sum_{n=1}^{\infty} p_{ii}^{n} = \sum_{n=1}^{\infty} \sum_{k=0}^{n/2} \frac{n!}{k!\,k!\,(n-2k)!} pa^k\, pb^k\, (1-pa-pb)^{n-2k}$$

when $\lambda = \mu$ $\quad pa = pb = \lambda(1-\lambda)$ $\quad pc = (1-\lambda)^2 + \lambda^2$, $p_{ii}^{n} \sim C n^{-1/2}$.

$$\sum_{n=1}^{\infty} p_{ii}^{n} = \sum_{n=1}^{\infty} n^{-1/2} = \infty \quad \text{recurrent}$$

when $\lambda < \mu$: $\quad p_{ii}^{n} \sim C$.